Commutative Algebra Problem Set #1

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Problem (Exercise 1)**.** *Let $\mathfrak{p}$ and $\mathfrak{q}$ be two prime ideals in a ring A such that neither $\mathfrak{p} \subseteq \mathfrak{q}$ nor $\mathfrak{q} \subseteq \mathfrak{p}$. Show that the ideal $\mathfrak{p} \cap \mathfrak{q}$ is not prime.*

Solution. To show that $\mathfrak{p} \cap \mathfrak{q}$ is not prime, it suffices to pick two a, b such that $ab \in \mathfrak{p} \cap \mathfrak{q}$, but $a, b \notin \mathfrak{p} \cap \mathfrak{q}$. Since neither $\mathfrak{p} \subseteq \mathfrak{q}$ nor $\mathfrak{q} \subseteq \mathfrak{p}$ holds, there exist elements $a \in \mathfrak{p} \setminus \mathfrak{q}$ and $b \in \mathfrak{q} \setminus \mathfrak{p}$. Notice that both elements are not in $\mathfrak{p} \cap \mathfrak{q}$, but their product is: on one hand, $ab \in \mathfrak{p}$ since $a \in \mathfrak{p}$ and $\mathfrak{p}$ is an ideal. Similarly, $ab \in \mathfrak{q}$ since $b \in \mathfrak{q}$. Thus, since $ab \in \mathfrak{p}$ and $ab \in \mathfrak{q}$, we conclude $ab \in \mathfrak{p} \cap \mathfrak{q}$. □

Problem (Exercise 2)**.** *A ring A is Boolean if $x^2 = x$ for all $x \in A$. In a Boolean ring A, show that:*

(i) $2x = 0$ *for all* $x \in A$*;*

(ii) every prime ideal $\mathfrak{p}$ is maximal, and $A/\mathfrak{p}$ is a field with two elements;

(iii) every finitely generated ideal in A is principal.

Solution.

Part (i). For arbitrary $x \in A$, we conduct the following steps:

$$2x = 4x - 2x = 4x^2 - 2x = (2x)^2 - 2x = 2x - 2x = 0$$

Part (ii). Take arbitrary $[x] \in A/\mathfrak{p}$ (we denote by $[\cdot]$ elements of quotient ring), where $\mathfrak{p}$ is a prime ideal. Note that $[x]^2 = [x^2] = [x]$, thus we have $[x]^2 - [x] = ([x] - [1])[x] = [0]$. Since $\mathfrak{p}$ is prime, $A/\mathfrak{p}$ is a domain, so either $[x] = [1]$ or $[x] = [0]$. This proves that $A/\mathfrak{p} = \{[0], [1]\}$ is a domain with two elements. However, it is clearly a field: the only non-zero element is $[1]$ and its inverse is simply $[1]$. Therefore, $A/\mathfrak{p} \cong \mathbb{F}_2$.

Part (iii). It suffices to prove this fact for an ideal generated by two elements: say, $\mathfrak{m} = \langle a_1, a_2 \rangle$. Namely, we prove that $\langle a_1, a_2 \rangle = \langle a \rangle$ for $a = a_1 + a_2 - a_1 a_2$. The general case will follow by applying induction.

$a \in \langle a_1, a_2 \rangle$. This inclusion is evident: a is a linear combination of a_1 and a_2.

$a_1, a_2 \in \langle a \rangle$. Note that

$$aa_1 = (a_1 + a_2 - a_1 a_2)a_1 = a_1^2 + a_2 a_1 - a_1^2 a_2 = a_1$$

Thus $a_1 \in \langle a \rangle$. Similarly, $a_2 \in \langle a \rangle$. Thus we have shown $\langle a_1, a_2 \rangle = \langle a \rangle$. □

Problem (Exercise 3)**.** *This exercises consists of two subproblems:*

(i) Show that a ring A is a domain if and only if it admits an injective homomorphism $A \hookrightarrow K$ into a field K.

(ii) Show that A is reduced (has no non-zero nilpotent elements) if and only if it admits an injective homomorphism $A \hookrightarrow \prod_{\alpha \in I} K_\alpha$ into a product of (possibly infinitely many) fields.

Solution.

Part (i). ($\Rightarrow$) Suppose A is a domain. Now consider the natural inclusion morphism $\iota : A \hookrightarrow \mathrm{Frac}(A), a \mapsto a/1$. Clearly, this is an injective homomorphism, and $K = \mathrm{Frac}(A)$ is indeed a field.

($\Leftarrow$) Suppose K is a field, and one is provided with the injective homomorphism $\gamma : A \hookrightarrow K$. We shall show that A is a domain. Take two elements $x, y \in A$ such that $xy = 0$. Take homomorphism of both sides: $\gamma(x)\gamma(y) = \gamma(0)$. Clearly, $\gamma(0) = 0$, so we are left with $\gamma(x)\gamma(y) = 0$. Since K is a field (and an integral domain, in particular), either $\gamma(x) = 0$ or $\gamma(y) = 0$. But since γ is injective, this implies that either $x = 0$ or $y = 0$. Thus A is a domain. □

Part (ii). ($\Rightarrow$) Recall that $\mathrm{Nil}(A) = \bigcap_{\mathfrak{p} \text{ prime}} \mathfrak{p}$. Since A is reduced, $\mathrm{Nil}(A) = 0$, so no two prime ideals intersect besides 0. This gives the natural embedding $A \hookrightarrow \prod_{\alpha \in I} A/\mathfrak{p}$ by sending $x \in A$ to the corresponding quotient $A/\mathfrak{p}$. But note that since $\mathfrak{p}$ is prime, $A/\mathfrak{p}$ is a domain, and thus we have a natural embedding into $\mathrm{Frac}(A/\mathfrak{p})$. Composing both embeddings produce the desired result.

($\Leftarrow$) Suppose we are given injective homomorphism $\varphi : A \hookrightarrow \prod_{\alpha \in I} K_\alpha$. Let $a \in A$ be nilpotent, so $a^n = 0$ for some n. Apply φ to obtain $\varphi(a^n) = \varphi(a)^n = 0$. This means that $\varphi(a)_\alpha^n = 0$ for each $\alpha \in I$. Since field has no non-zero nilpotents, we conclude that $\varphi(a)_\alpha = 0$ or simply $\varphi(a) = 0$. Since φ is injective, $a = 0$. Thus, there are no non-zero nilpotents in A, so A is reduced. □

Problem (Exercise 4)**.** *Let X be a compact Hausdorff space and let $A = C(X, \mathbb{R})$ be the ring of continuous functions on X. For $x \in X$, let $\mathfrak{m}_x \subseteq A$ be the set of all $f \in A$ such that $f(x) = 0$. Show that $\mathfrak{m}_x$ is a maximal ideal in A, and that every maximal ideal $\mathfrak{m} \subseteq A$ is of the form $\mathfrak{m}_x$ for a unique $x \in X$.*

Solution. To prove $\mathfrak{m}_x$ is a maximal ideal, we prove that $A/\mathfrak{m}_x$ is a field. I see two approaches how to prove this.

Approach #1 (mechanical). Take any $[f] \in A/\mathfrak{m}_x$. If $f \in \mathfrak{m}_x$ (that is, $f(x) = 0$), then $[f] = [0]$. Otherwise, the inverse of $[f]$ is always given by the constant function representative $[1/f(x)]$. This function exists since $f(x) \neq 0$, and moreover $[f][1/f(x)] = [f/f(x)] = [1]$.

Approach #2. Consider the evaluation morphism $\mathrm{ev}_x : A \to \mathbb{R}$ that sends

$f \mapsto f(x)$. Note that $\ker(\mathrm{ev}_x) = \mathfrak{m}_x$ by definition of the ideal $\mathfrak{m}_x$. Moreover, clearly $\mathrm{im}(\mathrm{ev}_x) = \mathbb{R}$ (since for any $\alpha \in \mathbb{R}$ we can find some continuous function $f \in A$ such that $f(x) = \alpha$). Thus, we induce the isomorphism $A/\mathfrak{m}_x \cong \mathbb{R}$. Since $\mathbb{R}$ is a field, $A/\mathfrak{m}_x$ is a field. □

Now we prove that all maximal ideals are of form $\mathfrak{m}_x$ with $x \in X$. For this, we will show the auxiliary proposition.

Proposition. *Let $\mathfrak{m}$ be a maximal ideal and $V(\mathfrak{m}) := \{x \in X : f(x) = 0 \text{ for all } f \in \mathfrak{m}\}$. Then, $V(\mathfrak{m}) = \{x\}$ for some $x \in X$, and $\mathfrak{m} \subseteq \mathfrak{m}_x$.*

Proof. We first show that this set is non-empty. Indeed, suppose otherwise: $V(\mathfrak{m}) = \emptyset$, thus for each $x \in X$ we can find $f_x \in \mathfrak{m}$ such that $f_x(x) \neq 0$. Thus, by continuity, these exists the neighbourhood U_x around x such that $f_x(U_x) \neq \{0\}$. Note that $\bigcup_{x \in X} U_x$ covers the whole X. Since X is a compact space, we might choose the finite subcovering: for instance, let $\bigcup_{i=1}^{n} U_{x_i} = X$. Let $h := \sum_{i=1}^{n} f_{x_i}^2$. Clearly, $h \in \mathfrak{m}$ since every $f_{x_i} \in \mathfrak{m}$. However, note that h is a strictly positive function over X since it is a sum of squares of strictly positive functions over corresponding open sets U_{x_i}. Therefore, $h \in \mathfrak{m}$ is a unit. However, the maximal ideal cannot contain a unit (otherwise $\mathfrak{m}$ is the whole ring). Contradiction.

Now suppose $x \in V(\mathfrak{m})$. Notice that $\mathfrak{m} \subseteq \mathfrak{m}_x$. Indeed, by definition of $V(\mathfrak{m})$, every element of $f \in \mathfrak{m}$ satisfies $f(x) = 0$, so we have the corresponding inclusion. But since $\mathfrak{m}$ is maximal, either $\mathfrak{m}_x = \mathfrak{m}$ or $\mathfrak{m}_x = A$. The latter is clearly not the case, so we conclude $\mathfrak{m}_x = \mathfrak{m}$.

Finally, we show uniqueness. Suppose $x, y \in V(\mathfrak{m})$, so in particular we have $\mathfrak{m}_x = \mathfrak{m}_y = \mathfrak{m}$. Now we apply the Tietze's extension theorem: define function $g : \{x, y\} \to \mathbb{R}$ such that $g(x) = 0, g(y) = 1$. Then, there exists the continuous extension $\widetilde{g} \in A$ such that $\widetilde{g}(x) = 0$ and $\widetilde{g}(y) = 1$. Since $\widetilde{g}(x) = 0$, $\widetilde{g} \in \mathfrak{m}_x$, but that implies $\widetilde{g} \in \mathfrak{m}_y$. But since $\widetilde{g}(y) = 1$, this is a contradiction.

Notice that this proposition finishes the theorem off. □

Problem (Exercise 5)**.** *Let again X be a compact Hausdorff space and let $A = C(X, \mathbb{R})$ be the ring of continuous functions on X. Show that every prime ideal in A is contained in a unique maximal ideal.*

Solution. We apply the same idea as in the previous proof. For prime ideal $\mathfrak{p}$, define the vanishing set $V(\mathfrak{p}) = \{x \in X : f(x) = 0 \text{ for all } f \in \mathfrak{p}\}$. First note that this set is non-empty for the same very reasons, as in the proposition above (unit cannot be contained in prime ideal either). Thus, there is some $x \in V(\mathfrak{p})$ and $\mathfrak{p} \subseteq \mathfrak{m}_x$ by the definition of $V(\mathfrak{p})$. It is remained to show that x is unique. To show that, we will need the following proposition.

Proposition. *For $E \subseteq X$, define $I(E) := \{f \in A : f(x) = 0$ for all $x \in E\}$. Then, $I(V(\mathfrak{a})) = \sqrt{\mathfrak{a}}$ for any ideal $\mathfrak{a} \subseteq A$.*

For now, assume that this proposition holds (we will prove it below shortly). Then, assume $x, y \in V(\mathfrak{p})$ and $x \neq y$. Then, by Tietze's extension theorem, we can find two functions $f, g \in A$ such that $f(x) = 0$, $f(y) = 1$, and $g(x) = 1$, $g(y) = 0$. Moreover, it is easy to check that we can construct f and g in such a way that $f(z)g(z) = 0$ for every $z \in X$. In particular, $fg \in I(V(\mathfrak{p}))$, thus $fg \in \sqrt{\mathfrak{p}} = \mathfrak{p}$. Thus, either $f \in \mathfrak{p}$ or $g \in \mathfrak{p}$. But f and g clearly do not vanish on $V(\mathfrak{p})$, so we get the contradiction. Thus, $V(\mathfrak{p}) = \{x\}$ for some $x \in X$, which proves the theorem. □

Proposition proof. $\sqrt{\mathfrak{a}} \subseteq I(V(\mathfrak{a}))$ is obvious: take $f \in A$ and we know that $f^n \in \mathfrak{a}$ for some n. For any $x \in V(\mathfrak{a})$ we know that $f^n(x) = 0$ by definition of $V(\mathfrak{a})$. Clearly, this implies $f(x) = 0$ and thus $f \in I(V(\mathfrak{a}))$.

The hard part is showing $I(V(\mathfrak{a})) \subseteq \sqrt{\mathfrak{a}}$. In other words, suppose f vanishes over all $x \in V(\mathfrak{a})$, then we shall show that $f^n \in \mathfrak{a}$ for some n. Define the set $Y := X \setminus V(\mathfrak{a})$. This set is open, because $V(\mathfrak{a}) = \bigcap_{f \in \mathfrak{a}} \{x \in X : f(x) = 0\}$ is the intersection of closed sets. Thus, for every $y \in Y$, find some function $f_y \in \mathfrak{a}$ and neighbourhood U_y such that $f_y(U_y) \neq \{0\}$. Clearly, $\{U_y\}_{y \in Y}$ is an open covering of Y, which means $\{U_y\}_{y \in Y} \cup V(\mathfrak{a})$ covers all X. This means that there exists a finite covering of Y: let it be $\{U_{y_i}\}_{i=1}^n$ with corresponding functions f_{y_i}. Now define $s := \sum_{i=1}^n f_{y_i}^2 \in \mathfrak{a}$. Note that we constructed s such that:

$$s|_{X \setminus V(\mathfrak{a})} > 0, \quad s|_{V(\mathfrak{a})} = 0, \quad s \in \mathfrak{a}.$$

Now define the following function:

$$\varphi(x) := \begin{cases} f(x)^2 / s(x), & \text{if } s(x) \neq 0, \\ 0, & \text{if } s(x) = 0. \end{cases}$$

Notice that $\varphi \in A$ and $f^2 = \varphi s$. Since $s \in \mathfrak{a}$, we conclude that $\varphi s \in \mathfrak{a}$ and thus $f^2 \in \mathfrak{a}$, so in particular $f \in \sqrt{\mathfrak{a}}$. □